

Ph.D. candidate in the Department of Computer Science, University of Virginia, Charlottesville, Virginia, United States. My research focuses on **human-computer interaction**, **human-AI interaction**, and **multimodal intelligent systems**. I design interactive agents that interpret human intent from multimodal input and enable transparent, steerable human-AI collaboration. My work spans agentic design systems, embodied AI, computer vision, teleoperation, XR, and haptics.

EXPERIENCE

Internship, Human-AI Co-Design

Autodesk | Future AI Team

May 2026 — Now

San Francisco, CA

- Building a real-time human-AI co-design system where designers and AI agents concurrently explore a shared design.
- Developing multi-agent orchestration for multimodal input understanding, design-space planning, and image generation.
- Designing memory-enabled agentic workflows that learn from designer feedback and support steerable collaboration.

Research Assistant, Interactive Systems

UVA Ultimate User Interface Lab | With Prof. Seongkook Heo

Sep 2020 — May 2026

Charlottesville, VA, USA

- Investigate how multimodal, embodied, and haptic interaction make intelligent and XR systems more effective and efficient.
- Design and implement end-to-end interactive systems combining computer vision, machine learning, and custom hardware.
- Lead empirical evaluation through experimental design, user studies, statistical analysis, and publication at CHI, ICRA, etc.

RESEARCH

VLA-Based Robotic-Hand Teleoperation with Haptic Feedback

Jan 2026 — Now

- Developing a VLA-based teleoperation system for multimodal intent understanding and dexterous robotic-hand control.
- Designing real-time haptic feedback and shared control to improve operator awareness and intervention in AI-assisted actions.

LLM-Enabled Mixed-Reality Interaction for Robot Mission Control

Feb 2025 — Mar 2026

- Built a mixed-reality interface that visualizes live robot sensing, localization, scene understanding, and control information.
- Enabled users to communicate with and supervise a robot via language and gesture for teleoperation and mission control.

Multimodal Tangible Interaction for Wearable Displays

Oct 2024 — Aug 2025

- Developed an interaction technique that turns uninstrumented everyday objects into tangible controls for wearable displays.
- Built a transformer-based perception pipeline for hand-object detection and intent inference, with 90.0% and 97.5% accuracy.

ML-Supported Self-Haptic Interaction in Extended Reality

Jan 2023 — Apr 2024

- Created an on-hand touch interface supported by a machine-learning model with 92% touch-detection accuracy.
- Designed coordinated hand-gaze interaction techniques for precise and comfortable multi-window productivity in XR.

Adaptive Shape Display for Visuo-Haptic Interaction in Virtual Reality

Feb 2022 — May 2023

- Built a foldable shape-changing device that uses shape approximation and visuo-haptic illusions to render virtual surfaces.
- Demonstrated improved realism, control, and 3D-modeling performance over tablet and mid-air input through a user study.

EDUCATION

University of Virginia, Master/Ph.D., Computer Science, USA, GPA: 3.89/4.00

Aug 2018 — Dec 2026

Zhejiang University, Bachelor, Computer Science, China, GPA: 3.70/4.00

Sep 2014 — Jun 2018

TECHNICAL SKILLS

AI Systems	Multi-agent, agentic AI, multimodal interaction, LLM/VLM, computer vision, PyTorch
Interactive Systems	XR, Unity, ROS, Arduino, OptiTrack, haptic systems, circuit/PCB design, 3D modeling
Programming	Python, C#, C/C++, Java, R, JavaScript, TypeScript, React, MATLAB

SELECTED PUBLICATIONS

[GOOGLE SCHOLAR](#)

1. **Ying, W.**, Kim, Y., Rahman, A., *et al.* Redirected Pinch: Efficient and Comfortable Bare-Hand Interaction for 2D Windows in VR. *Proceedings of the CHI Conference on Human Factors in Computing Systems* (April 13-17, 2026).
2. **Ying, W.** & Heo, S. Physical surfaces make touch interactions in virtual reality precise, efficient, and bimanual. *International Journal of Human-Computer Studies* (June 9, 2026).
3. **Ying, W.** & Heo, S. Enhancing VR Sketching with a Dynamic Shape Display (**Best Paper Honorable Mention**). *ACM Symposium on Virtual Reality Software and Technology* (October 9-11, 2024).
4. Hildebrandt, C., **Ying, W.**, Heo, S., *et al.* Mimicking Real Forces on a Drone Through a Haptic Suit to Enable Cost-Effective Validation. *IEEE International Conference on Robotics and Automation* (May 29-June 2, 2023).
5. Hu, E., Grønbaek, J. E. S., **Ying, W.**, *et al.* ThingShare: Ad-Hoc Digital Copies of Physical Objects for Sharing Things in Video Meetings. *Proceedings of the CHI Conference on Human Factors in Computing Systems* (April 23-28, 2023).